25900356 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCECT203 - INTEGRATED ELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. For an NPN BJT in the cut-off mode, what are the states of the Emitter-Base (EB) and Collector-Base (CB) junctions? [U] / [CO1]

   a). EB reverse-biased, CB forward-biased

   b). EB forward-biased, CB forward-biased

   c). EB reverse-biased, CB reverse-biased

   d). EB forward-biased, CB reverse-biased

2. Sequence of 2-bit up counter? [K] / [CO2]

   a). 00-01-11-10

   b). 00-01-10-11

   c). 11-10-01-00

   d). 10-11-01-00

3. Which unit is used to express slew rate? [K] / [CO3]

   a). V

   b). V/ms

   c). V/μs

   d). V/s

4. Input offset voltage of an Op-amp is : [K] / [CO3]

a). The voltage required between inverting and non-inverting inputs to make output zero

b). The minimum input voltage required to start amplification

c). The difference between bias currents of the input terminals

d). The maximum voltage that an Op-amp can amplify

5. Which of the following IC is commonly used as a general-purpose operational amplifier? [K] / [CO3]

a). IC 555

b). IC 741

c). IC 7805

d). IC 3525

6. A BJT has how many layers and junctions? [K] / [CO1]

a). 2 layers, 1 junction

b). 3 layers, 2 junctions

c). 4 layers, 3 junctions

d). 2 layers, 2 junctions

7. Johnson counter is also called? [K] / [CO2]

a). Twisted ring

b). Ripple

c). Binary

d). Modulo

8. The NE555 timer can operate in: [K] / [CO3]

a). Monostable mode

b). Astable mode

c). Bistable mode

d). All of the above

9. The closed-loop gain of a negative feedback amplifier is generally: [U] / [CO1]

a). Greater than the open-loop gain

b). Less than the open-loop gain

c). Equal to the open-loop gain

d). Unaffected by feedback

10. Synchronous decade counters avoid? [U] / [CO2]

a). Noise

b). Propagation delay

c). Clock pulses

d). Reset

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Define a Single-stage CE Amplifier and sketch its circuit with voltage divider biasing. [K] / [CO1]
12. Draw the block diagram of a Parallel-In Serial-Out (PISO) shift register and explain the role of the control input (shift/load). [K] / [CO2]
13. List and briefly explain any two op-amp open-loop configurations. [U] / [CO3]
14. Explain the principle of RC phase shift oscillator using Op-Amp. [U] / [CO3]
15. Explain the role of timing resistor and capacitor in IC 566. [U] / [CO3]
16. Discuss the essentials of an oscillator. [K] / [CO1]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Draw and explain the dc equivalent circuit and dc load of a single stage CE amplifier. [K] / [CO1]
18. Design a 4-bit Serial-In Parallel-Out (SIPO) register using D flip-flops. Show the output sequence for the input 1011. [K] / [CO2]
19. Derive the expression for the voltage gain of a closed-loop non-inverting amplifier. [U] / [CO3]
20. With the circuit diagram and derivation, explain the working of a differentiator. [U] / [CO3]

[10x2 = 20]